

Daniel Burger

EXPERIMENT

GRANT ALLOCATED \$10,000

One-Sentence Description

Brain-in-vat with modern perfusion technology, bringing back forgotten research on sustaining living brains outside the body.

Proposed Project

Daniel is resurrecting and modernizing the old research programme on isolated brain perfusion: keeping a brain alive and functional outside the body (“brain-in-vat”).

A part of his project goes into literature review on Soviet experiments: primary sources from Brukhonenko’s 1928 publications and related untranslated Russian literature.

Daniel’s near-term experiment is to slowly kill the body of a pig while keeping the brain alive, but replacing the heart with external life support, thereby achieving the result of a living brain in a dead body. This is the closest to “brain-in-vat” without physically removing the brain out of the body. The three-stage explanation involves: baseline ECoG recording and transition to artificial heart-lung support, gradual vascular isolation of the brain, then full anatomical separation into a perfusion chamber.

Experimental results will be published as a paper.

Revival Relevance

Daniel’s experimental work is reviving a discontinued research program with modern technology. His project coheres with the revival theme the most, as his work sits as a reinvented extension of a lineage:

- 1925, the Soviet physiologist Sergei Brukhonenko kept isolated dog heads alive using an early heart-lung machine he called “autojektor”;
- 1950s, Vladimir Demikhov did head-grafting experiments on dogs;
- 1960s-70s, Robert White isolated a primate brain under cooling and performed a monkey head transplant, with the animal retaining cranial nerve function, normal EEG, and senses for several days.

The old program, which ran roughly from 1920s-1971 in the USSR and the US, stalled due to a confluence of factors: rising animal-rights movement, shifting funding priorities, absence of immediate downstream applications, technical dead end (spinal cord failure leading to paralysis), and scholarly consensus in shifting direction out of head transplantation.

Now is a good time for a reinvention of the old program, as several obstacles have been lifted:

1. Organ perfusion is a mature commercial industry
2. Recent post-mortem perfusion papers showed more confidence in brain tissue’s resilience
3. Culturally, BCIs and longevity research, alongside organoid work, are becoming more normalized

A part of Daniel’s proposed project is also to translate primary sources from Soviet scientific literature.

Wendi's Notes

1. Daniel has a strong expertise to tackle this problem.

With expertise in brain-computer interfaces, neural tissue engineering, and computational neuroscience, Daniel is the founder of [EightSix Science](#). The company focuses on gradual brain replacement with engineered biohybrid tissues, and this stream of work can also contribute significantly to building ultra high bandwidth brain-computer interfaces that conventional electrodes cannot achieve.

Jenn and Chloe, without knowing Daniel also applied to Revival, mentioned Daniel as a thinker and experimentalist they admire, in the Revival Fund application. They cited Daniel's 2024 work as [the first to deliver stimulation to brain organoids in the lab via remote web interface](#).

Daniel recently spoke at the [Machine Consciousness Conference](#) and [Vitalist Bay](#) (Longevity) in Berkeley, alongside Max Hodak, Joscha Bach, David Pearce, Anders Sandberg, and many other influential thinkers in longevity, BCI, artificial life, etc.

2. Daniel has already gathered the rest of the resources needed for experimentation. Funding is the last piece of the puzzle.

Daniel's company has procured the animal protocol and are technically ready to run the experiments. In the two months before submission and our interview call, they gained more confidence from a trial run, and projected a successful experiment by the end of this year. They have collaborated with researchers in Japan on this front as well.

3. Big-if-True: Revival funding unlocks the first stage proof-of-concept for brain replacement.

Daniel's proposal for Revival constitutes a core foundational investigation within the bigger goal of achieving brain replacement. His conviction is that brain replacement might never work if we don't have a brain-in-vat first. A brain-in-vat can be continuously imaged with MRI, operated on without anaesthetic and used for drug testing, making it a more controllable research.

Daniel's current experiments involve small grafts into the cortex of two pigs, running histology to see how the new tissue functions, with electrodes placed before and after implantation.

Daniel outlines 5 stages of achieving brain isolation:

1. Circulatory Independence
2. Hepatic/Renal Support
3. Head Isolation (C1 Transection)
4. Skull Cap Replacement
5. Full Ex Cranio (Brain Outside)

What the Revival Funding enables, is a Stage 1 experiment that helps Daniel and his team get 7-figure funding for the next stages. Given the old image tied to this stream of research, brain replacement is still underfunded, and Daniel hopes to use the successful experimental result from Revival as a major first validation, showing both the efficacy and ethics of his approach, thereby unlocking more standard science grants.

This is exactly what Revival Fund was built for: seeding a revival kickoff on a discontinued research lineage.